DTC P2579:00 [SKYACTIV-D 1.5]

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DTC P2579:00	Speed sensor assy. (turbocharger) range/performance problem
DETECTION CONDITION	• When the following conditions are met, the PCM detects **5 times** that the turbine speed detected by the speed sensor component (turbocharger with variable turbine geometry) is **lower than 17,000 rpm or more** compared to the turbine speed estimated by the PCM. **MONITORING CONDITIONS** — Gear position: 3rd or 5th — Vehicle speed: **30—110 km/h {19.0—68.3 mph}** — Intake air temperature: **20—35 °C {68—95 °F}** — APP: **30—50 %** — Engine coolant temperature: **80 °C {176 °F} or more** — Barometric pressure: **98 kPa {736 mmHg, 29 inHg} or more** — DPF regeneration control is not performed **Diagnostic support note** • This is a continuous monitor (Turbo system). • The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle. • FREEZE FRAME DATA/Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	• PCM restricts engine torque.
POSSIBLE CAUSE	• Speed sensor assy. (turbocharger) connector or terminals malfunction • PCM connector or terminals malfunction • Speed sensor assy. (turbocharger) malfunction • Turbocharger malfunction (turbine wheel and/or compressor wheel damaged, stuck) • PCM malfunction
SYSTEM WIRING DIAGRAM	• Not applicable

Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS WHEN DTC WAS DETECTED TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture of the PC function. • Record the freeze frame data/snap shot data.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**VERIFY RELATED PENDING CODE AND/OR DTC** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any other PENDING CODEs and/or DTCs present?	Yes	Go to the applicable PENDING CODE or DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.
4	**INSPECT SPEED SENSOR ASSY. (TURBOCHARGER) CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the speed sensor assy. (turbocharger) connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 7.
		No	Go to the next step.

STEP	INSPECTION		ACTION
5	**INSPECT INSTALLATION OF SPEED SENSOR ASSY. (TURBOCHARGER)** • Inspect installation of speed sensor assy. (turbocharger). (See SPEED SENSOR ASSY (TURBOCHARGER WITH VARIABLE TURBINE GEOMETRY) REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) • Is the speed sensor assy. (turbocharger) installed securely?	Yes	Go to the next step.
		No	Retighten the speed sensor assy. (turbocharger), then go to Step 7. (See SPEED SENSOR ASSY (TURBOCHARGER WITH VARIABLE TURBINE GEOMETRY) REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
6	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to the next step.
		No	Replace the turbocharger, then go to the next step. (See TURBOCHARGER WITH VARIABLE TURBINE GEOMETRY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
7	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Perform the KOER self test. (See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
8	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.